INFOSYS SPRINGBOARD

INTERNSHIP PROJECT DOCUMENTATION

PROJECT NAME

PlantDocBot: AI Plant Disease Diagnosis via Chat, Image and Voice

DEVELOPED BY

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TECHNOLOGIES USED

Python | PyTorch | Deep Learning | ResNet50 | Streamlit | OpenAI Whisper | Groq API

INPUT MODES

Image | Chat | Voice

INFOSYS SPRINGBOARD INTERNSHIP

AI-powered assistance for faster and accessible plant disease diagnosis

1.Introduction

Agriculture plays an important role in the economy and livelihood of millions of people, particularly in countries like India. Plant diseases are one of the major challenges in agriculture because they can affect crop health, quality, and overall production. Early identification of diseases is important to prevent their spread and reduce crop losses.

Traditionally, plant diseases are identified by visually observing symptoms on leaves, stems, or fruits. This method depends largely on human experience and agricultural knowledge. In many situations, accurate identification can be difficult, especially when similar symptoms are caused by different diseases. Farmers in rural areas may also have limited access to agricultural experts for immediate guidance.

PlantDocBot: AI Plant Disease Diagnosis via Chat, Image and Voice is developed as an AI-based solution to assist users in plant disease diagnosis. The system combines **Artificial Intelligence, Deep Learning, Computer Vision, and Natural Language Processing** to provide an interactive and accessible diagnosis platform.

For image-based diagnosis, the system uses the **ResNet50 deep learning model** to analyze plant leaf images and predict the corresponding disease. In addition to image input, users can interact with the system through **Chat/Text and Voice inputs**, making the application more flexible and user-friendly.

The system also provides useful information about the identified disease, including its **name, symptoms, causes, treatment, and prevention methods**. This allows users to better understand the detected problem and take appropriate action.

2.Problem Statement

Plant disease identification is often a time-consuming and challenging task when performed manually. Incorrect or delayed identification can result in improper treatment, disease spread, crop damage, and financial losses for farmers.

Another major challenge is the limited availability of agricultural experts, particularly in rural and remote areas. Users may not always have access to timely and reliable information about plant diseases. Furthermore, depending on only one method of interaction may make a diagnosis system less accessible to different types of users.

Therefore, there is a need for an **AI-powered, easy-to-use, and multi-input plant disease diagnosis system** that can assist users in identifying diseases quickly and provide relevant information. PlantDocBot addresses this problem by supporting **Image, Chat/Text, and Voice inputs** and using deep learning-based image classification for plant disease detection.

3.Project Objectives

The main objective of **PlantDocBot** is to develop an AI-powered plant disease diagnosis system that helps users identify plant diseases quickly and conveniently.

• To detect plant diseases from plant leaf images using Deep Learning.
• To use the ResNet50 model for plant disease classification.
• To provide multiple interaction methods through Image, Chat/Text, and Voice inputs.
• To display a confidence score along with the predicted disease.
• To provide useful information about diseases, including symptoms, causes, treatment, and prevention.
• To provide a simple and user-friendly system for both technical and non-technical users.

4. Key Features

• **Image-Based Disease Detection:** Users can upload a plant leaf image, and the trained ResNet50 model analyzes the image to predict the possible disease along with its prediction confidence score.

• **Chat/Text Interaction:** Users can enter questions or describe plant-related symptoms through text and receive relevant information from the system.

• **Voice Input:** Users can provide their queries using voice. The voice input is converted into text and processed by the system.

• **AI-Powered Assistance:** The system combines Artificial Intelligence and Deep Learning to provide intelligent assistance for plant disease diagnosis.

• **Disease Information:** The system provides useful information such as disease name, symptoms, causes, treatment, and prevention methods.

• **Quick and User-Friendly Diagnosis:** The application provides a simple interface and quick responses, making preliminary plant disease identification convenient and accessible.

5. Technology Stack

The PlantDocBot system is developed using a combination of Artificial Intelligence, Deep Learning, Computer Vision, and web application technologies.

• **Python** – Used as the primary programming language for application development and model integration.

• **PyTorch** – Used for developing and running the deep learning model.

• **ResNet50** – Used for plant leaf image classification and disease prediction.

• **Deep Learning** – Used to learn image patterns and identify plant disease classes.

• **Torchvision** – Used for image transformations and computer vision-related operations.

• **Streamlit** – Used to develop the interactive web-based user interface.

• **OpenAI Whisper** – Used for converting voice input into text.

• **Groq API** – Used for AI-powered processing and responses.

• **FastAPI** – Used for backend API services and communication.

• **NumPy** – Used for numerical operations and data processing.

6. System Workflow

The overall working process of PlantDocBot can be summarized as follows:

User Input → Input Processing → AI/Deep Learning Model → Disease Analysis → Prediction & Information

The system accepts **Image, Chat/Text, or Voice** input from the user. For image diagnosis, the uploaded plant image is preprocessed and passed to the trained **ResNet50 model**, which predicts the disease and provides a confidence score. Text and voice inputs are processed to understand the user's query and generate relevant plant disease information.

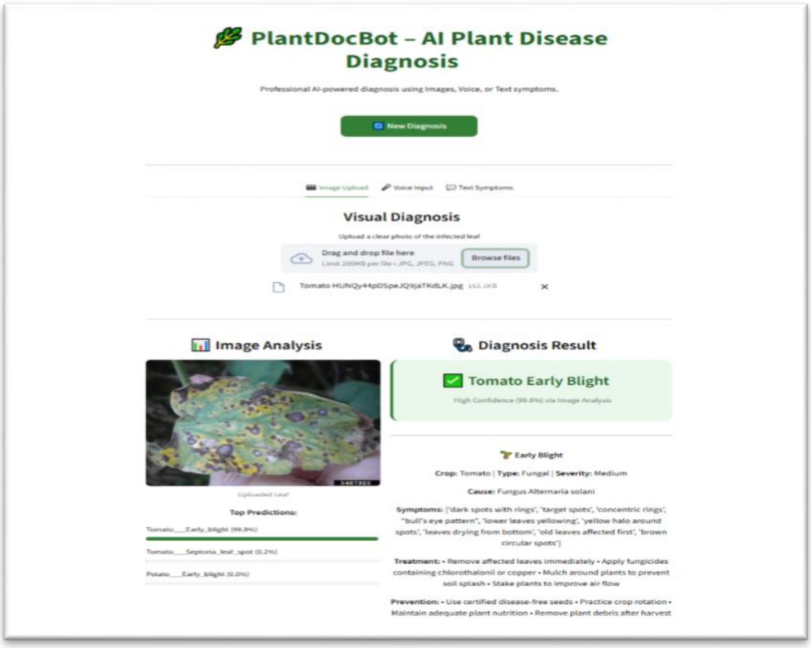
7. User Input & Diagnosis Modes

PlantDocBot provides three input modes that allow users to interact with the system and receive plant disease-related information.

1.Image Input

Users can upload a **plant leaf image** through the application. The uploaded image is immediately processed by the trained **ResNet50 Deep Learning model**. The system predicts the possible disease and displays the **confidence score** along with relevant disease information.

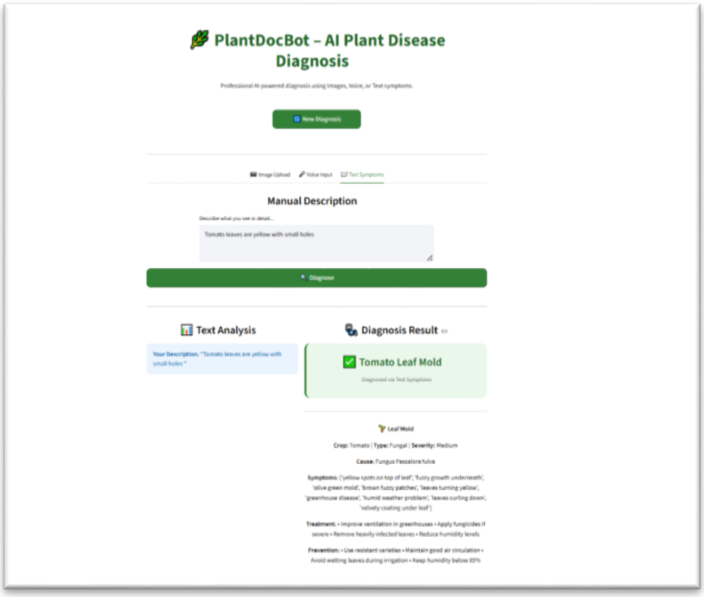
Image Diagnosis Flow:
Upload Image → Image Processing → ResNet50 Model → Disease Prediction → Confidence Score



2. Chat Input

Users can enter their questions or describe plant-related symptoms through the **Chat/Text interface**. The system processes the provided text and generates a relevant response to assist the user with plant disease-related queries.

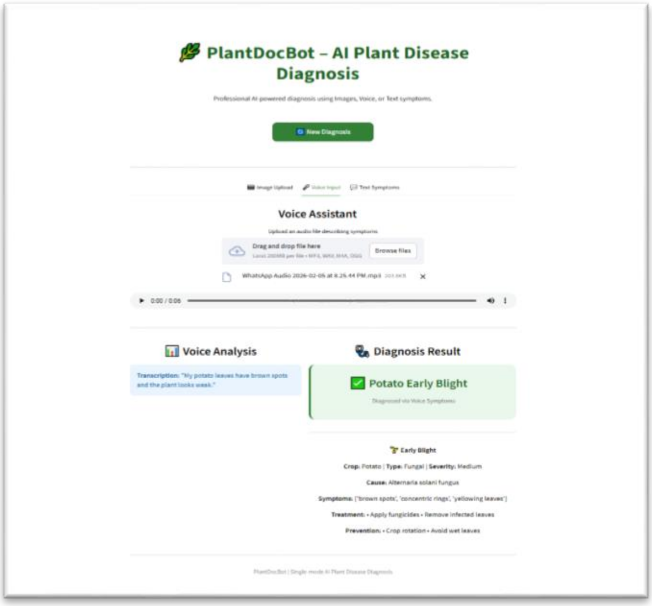
Chat/Text Flow:
Enter Text → Query Processing → AI Analysis → Response



3.Voice Input

Users can provide voice-based queries by **uploading an audio file** through the application. The uploaded audio is processed using **speech-to-text technology**, which converts the spoken content into text. The converted text is then processed by the system to generate a relevant response.

Voice Flow:
Upload Audio File → Speech-to-Text → Query Processing → AI Response



8.AI Model, Dataset & Results

1.AI Model

PlantDocBot uses **ResNet50**, a deep learning model based on Convolutional Neural Networks (CNNs), for plant disease image classification. The model learns visual patterns from plant leaf images and predicts the most likely disease class.

The model is implemented using **PyTorch** and is integrated into the application for real-time plant disease prediction.

2.Dataset

The model is trained using plant disease images from the **PlantVillage** and **PlantDoc** datasets.

• **PlantVillage Dataset** – Provides a large collection of plant leaf images representing different crops and disease categories.

• **PlantDoc Dataset** – Provides real-world plant leaf images and adds diversity to the training data.

These datasets help the model learn different visual characteristics of healthy and diseased plants.

3.Disease Prediction

When a user uploads a plant leaf image, the image is processed and passed to the trained **ResNet50 model**. The model predicts the most likely disease class based on the visual features present in the image.

The application also displays a **confidence score** along with the prediction, indicating the model's confidence in the identified disease.

4.Diagnosis Information

After identifying the disease, PlantDocBot provides useful information such as:

• Disease name
• Crop/plant type
• Symptoms
• Causes
• Treatment
• Prevention methods

9.Conclusion, Future Scope & Repository

1.Conclusion

PlantDocBot: AI Plant Disease Diagnosis via Chat, Image and Voice provides an AI-based approach to assist users in identifying plant diseases. The system combines **Deep Learning, Computer Vision, Speech Recognition, and AI-based interaction** to make plant disease diagnosis simple and accessible.

The **ResNet50** model is used for image-based disease prediction, while **Chat/Text and Voice** inputs provide additional ways for users to interact with the system. The application also provides useful disease-related information and confidence scores to support better understanding of the prediction.

Overall, PlantDocBot demonstrates how AI and Deep Learning can be applied to address a practical agricultural problem and provide quick preliminary assistance to users.

2.Future Scope

The system can be further improved by:

• Adding more crops and plant disease classes.
• Increasing the size and diversity of the training dataset.

• Improving model accuracy and prediction performance.
• Adding multilingual support for regional languages.
• Integrating real-time camera-based plant disease detection.
• Providing more advanced agricultural recommendations and expert assistance.

3.Project Repository

The complete source code, training notebooks, model files, application files, and project resources are available in the GitHub repository.

GitHub Repository:
PlantChatBot – AI Plant Disease Diagnosis System

https://github.com/RadhikaDPingale23/PlantChatBot-

Project Status

Completed — AI-based plant disease diagnosis system with Image, Chat, and Voice input support.